

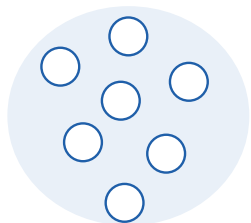
update every sample, then re-measure at the next step

4

$$G_i \propto [(y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F})] J_i^\top \nabla f_A(m_i)$$

servo the deviation gain

batch at time t

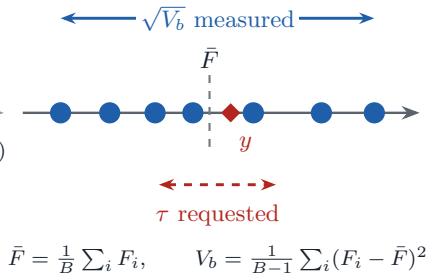


$$x_t^{(i)}, i = 1, \dots, B$$

1

$$F_i = f_A(m_i)$$

frozen guide



2

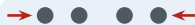
$$F$$

$$e = \frac{V_b - \tau^2}{\tau^2}$$

$$w_{\text{eff}} = 1 + \eta e$$

3

$e > 0$: contract



$e = 0$: plug-in

$e < 0$: expand



BDG

num_i =

$$\underbrace{(y - \bar{F})}$$

centring: gain w , fixed

—

$$\underbrace{w_{\text{eff}} (F_i - \bar{F})}$$

deviation: gain w_{eff} , servoed

zero extra NFE

nothing trained

knobs: $\eta, \tau = \tau_{\text{mult}} s$

$w_{\text{eff}} \geq 1 - \eta$